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AI Based Optimization for Shutdown Alignment for ADNOC Complex Value Chain: Development, Validation, and Deployment of VAST at DNOC

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Abstract

ADNOC manages a complex, multi-entity value chain with hundreds of process units and over 200 planned shutdowns each year. Coordinating these shutdowns to maintain asset integrity while maximizing value requires navigating an enormous, combinatorial search space. This paper presents the design, development, and deployment of the VCO ADNOC Shutdown Tool (VAST), an AI-enabled optimization application that aligns shutdowns across ADNOC's integrated network. VAST combines genetic search for shutdown-date selection with linear and non-linear optimal flow solvers in a hybrid architecture. We describe the modeling journey along with five algorithm versions that culminated in a parallelized, deterministic-assisted solution. User Acceptance Testing (UAT) covered 93 tests with 87 passes (94%), validating network models for upstream, gas, refining, and petrochemicals, and confirming reporting, flowsheet visualization, and API integration. Solver benchmarking identified HiGHS as the preferred LP engine for the full-network problem. Runtime scales with the number of enabled shutdowns (illustratively ~2 hours for ~72 and ~4 hours for ~175), motivating hardware and algorithmic speed-ups. We share lessons learned on model validation, repeatability, usability, and IT architecture, and outline a practical operating model for annual shutdown planning and sensitivity analysis. The resulting system has been deployed in production with SSO, role-based access, and documentation handed over, enabling ADNOC to run integrated shutdown-alignment exercises and scenario comparisons at enterprise scale.

Keywords: shutdown alignment, value chain optimization, genetic algorithms, linear programming, non-linear programming, hybrid optimization, UAT, flowsheet, ADNOC

Introduction

ADNOC operates a tightly interconnected upstream-midstream-downstream-petrochemicals network in which planned shutdowns are essential for safety, integrity, and growth. These shutdowns must be coordinated across units, plants, and subsidiaries so that product balances, constraints, and business objectives remain feasible and economically sound. Traditional methods struggle at this scale because the decision space—what to shut, when, and for how long, subject to operational and economic constraints

—explodes combinatorially. The VCO ADNOC Shutdown Tool (VAST) addresses this challenge by combining model-based optimal flow calculations with AI-enabled global search to recommend high-value, feasible shutdown schedules over multi-year horizons.

Statement of Theory and Definitions

Genetic Algorithms AI component

Genetic algorithms (GAs) are widely recognized as a core component of Artificial Intelligence (AI), particularly within the domain of evolutionary computation. Here's a clear explanation of why GAs was chosen as AI components, especially in the context of VAST:

Genetic algorithms are search heuristics inspired by the process of natural selection. They belong to a class of intelligent optimization techniques that mimic biological evolution to solve complex problems. Here's how they align with AI principles:

Learning and Adaptation

GA's evolves solutions over time by selecting, recombining, and mutating candidate solutions. This iterative process allows the algorithm to learn from previous generations and adapt to constraints and objectives—hallmarks of intelligent behavior.

Problem Solving in Complex Spaces

AI systems are designed to solve problems that are too complex for brute-force or deterministic methods. GAs excel in combinatorial optimization, where the solution space is vast and non-linear—like aligning shutdowns across ADNOC's multi-asset network in VAST.

Autonomous Decision-Making

GAs autonomously explore and evaluate billions of possible scenarios without human intervention. In VAST, the GA autonomously determines optimal shutdown sequences based on economic impact, operational constraints, and flow dynamics.

Heuristic Intelligence

Unlike traditional solvers, GAs do not require gradient information or convexity assumptions. They use heuristics to intelligently navigate the solution space, making them suitable for real-world, noisy, and multi-objective problems.

Integration with Other AI Techniques

In VAST, GAs is integrated with LP/NLP solvers and deterministic modules, forming a hybrid AI system. This multi-layered architecture reflects modern AI systems that combine learning, reasoning, and optimization.

In VAST, the genetic algorithm ([Figure 1](#)):

- Generates diverse shutdown plans (population).
- Evaluates each plan based on economic and operational metrics (fitness).
- Selects and recombines high-performing plans (crossover and mutation).
- Iteratively improves solutions until optimal shutdown alignment is achieved.
- This process mirrors intelligent planning and decision-making, qualifying GAs as a genuine AI component.

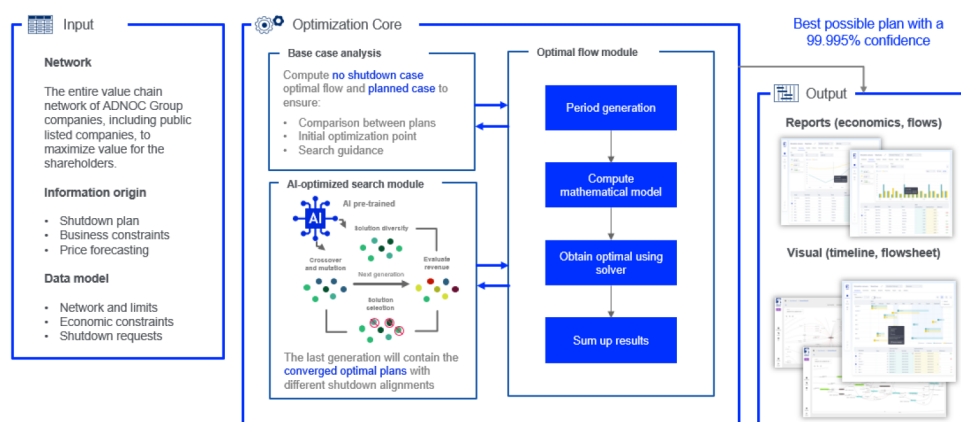


Figure 1—Overall login of VAST as a shutdown optimization solution

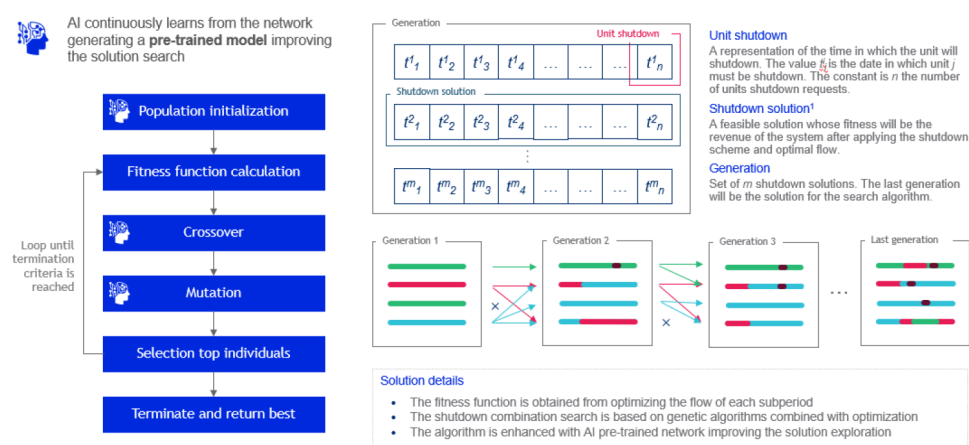


Figure 2—AI component and flow solver configuration

Optimization framing

We frame shutdown alignment as a constrained, mixed discrete-continuous optimization, which captures the inherent complexity of coordinating maintenance schedules across interconnected industrial networks. Discrete variables are used to encode shutdown window choices, including the start dates and durations for individual units, allowing for flexibility in scheduling decisions while adhering to predefined maintenance requirements. Continuous variables, on the other hand, represent network flows and compositions, ensuring that material balances, product specifications, and operational dependencies are accurately modeled throughout the system.

The objective function approximates total economic value (e.g., revenue net of transfer costs) over the planning horizon subject to operational constraints. This includes revenue from product sales, net transfer costs, while accounting for penalties, operational inefficiencies, and shutdown-related expenses. The formulation incorporates a wide range of operational constraints, such as capacity limits, blackout periods, inter-unit dependencies, and regulatory requirements, ensuring that the solution is both feasible and economically optimal.

Periods and subperiods

A "period" is a span during which exogenous inputs (prices, exogenous constraints) are stationary. A "subperiod" further partitions a period so that shutdown states remain constant; Optimal flows are recomputed for each subperiod to accurately capture the economic and operational impacts of any given shutdown combination. This hierarchical structure of periods and subperiods ensures that the optimization

framework can balance the stability of external inputs with the flexibility required to model dynamic shutdown scenarios effectively.

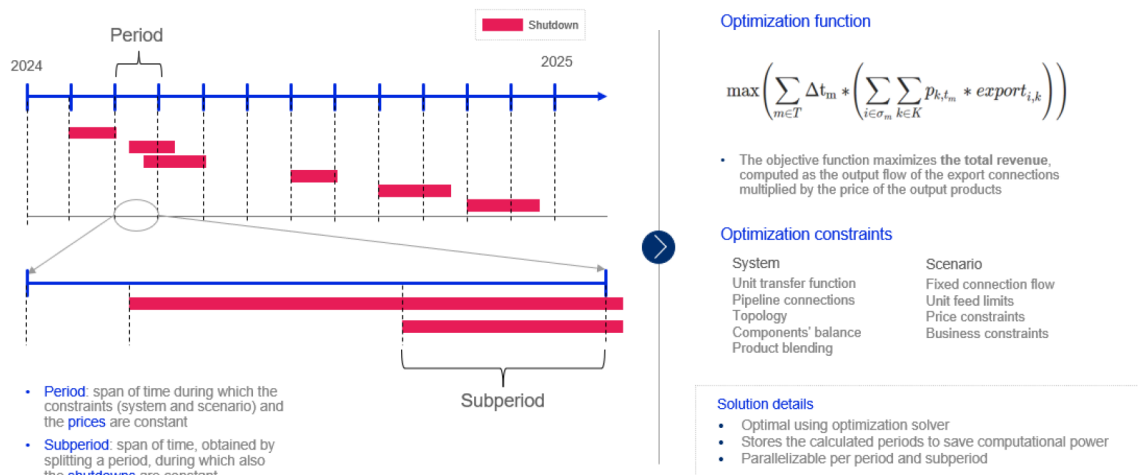


Figure 3—Periods, Subperiods and Optimization function

Solvers and search

VAST supports three flow-solver configurations: (i) pure non-linear programming (NLP) for compositionally rigorous units, (ii) pure linear programming (LP) via a linearized network, and (iii) a hybrid LP^NLP approach. These configurations are designed to balance computational efficiency with the need for rigorous modeling of operational constraints and economic impacts:

Pure Non-Linear Programming (NLP). This configuration is used for units that require compositionally rigorous modeling, such as those with complex composition processes or non-linear relationships between inputs and outputs. While computationally intensive, this approach ensures high fidelity in scenarios where precision is critical, such as optimizing flows in petrochemical units or gas processing facilities.

Pure Linear Programming (LP). The LP configuration leverages a linearized representation of the network, simplifying flow calculations to achieve faster computational performance. This approach is particularly useful for large-scale networks where the relationships between flows and constraints can be reasonably approximated as linear. LP is ideal for scenarios requiring rapid evaluations, such as initial broad searches over shutdown.

Hybrid LP→NLP Approach. The hybrid configuration combines the strengths of both LP and NLP, employing a two-stage process to balance speed and precision. The hybrid approach uses LP during the alignment phase to efficiently explore shutdown combinations across the network. Once the best alignment is identified, NLP is applied to evaluate and refine the solution, ensuring compositionally rigorous checks for critical subproblems.

To identify optimal shutdown combinations across the network, VAST employs a global search methodology based on a genetic algorithm. This evolutionary algorithm mimics natural selection processes, using crossover, mutation, and selection operators to explore the solution space effectively. The genetic algorithm is particularly well-suited for discrete optimization problems, such as determining the start dates and durations of shutdowns, where the solution space is vast and non-linear.

Description and Application of the VAST System

Modeling scope and evolution

LP vs NLP Shutdown Alignment Study. To validate that a linearized network adequately preserves the economic signal for shutdown alignment, we executed a two-part study on a representative gas network over a 30-day horizon. The design compared LP (linearized flow model) versus NLP (pool/composition model) by exhaustively evaluating daily start-date combinations for one and two shutdowns while holding exogenous constraints constant. This removed GA search from the loop, isolating the flow-solver effect on value and alignment.

Study setup. For each candidate start day (and for both units in the two-shutdown case), we computed economic outcomes with both LP and NLP and selected the best per method. Price curves varied by 5-day windows for key products (e.g., Polyethylene, Ruwais C3/C4) while fixed-flow purchases were held constant, ensuring a controlled but dynamic valuation backdrop. The LP evaluation per combination typically completed in ~1-3 seconds versus ~2-4 minutes for NLP, turning runs that would take hours/days into minutes when scanning many candidates.

Study 1 — Single shutdown (POC model NGL TR3, 5 days). We assessed ~60 feasible placements across the month. LP and NLP selected the same optimal start date, indicating that the linearized model captured the price-driven shape of the revenue curve sufficiently for date-choice decisions. In the reported best case, LP $\approx$ 2600.55 M\$ vs NLP $\approx$ 2530.97 M\$, while the rank-ordering by date coincided across methods—crucial for GA's convergence when LP is used for exploration.

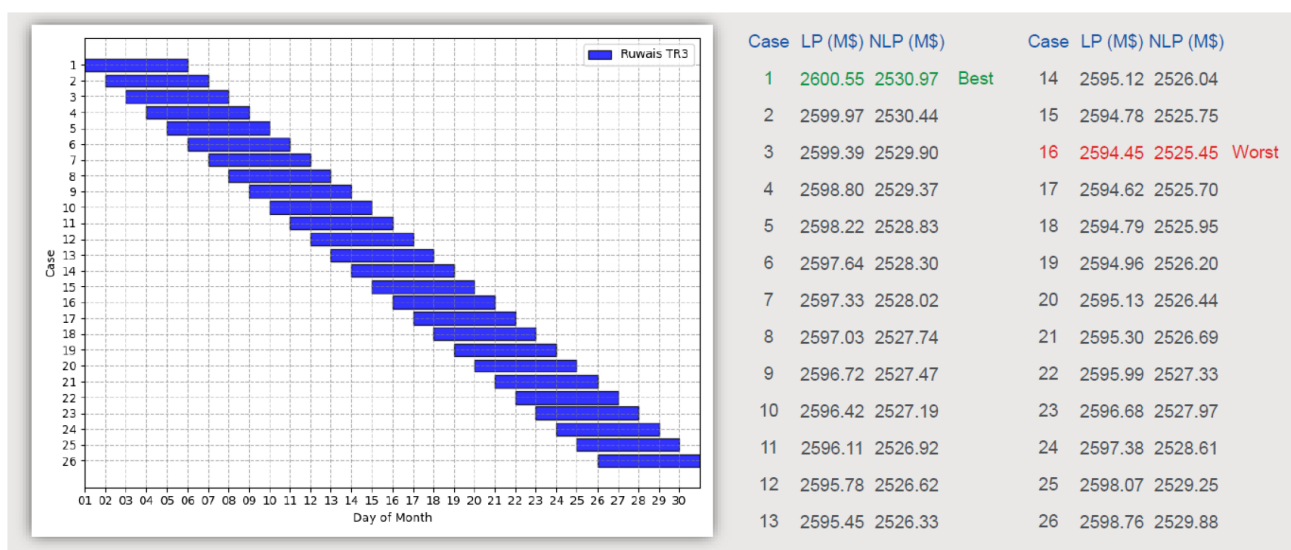


Figure 4—LP vs NLP on fixed single shutdown evaluation

Study 2 — Two shutdowns (POC model NGL TR3, 5 days; POC model Cracker 3, 3 days). We evaluated ~1,350 combinations. LP and NLP best solutions reflected the same alignment pair(s) of start dates, and the revenue surfaces showed very similar shapes, meaning LP's linearization preserved the relative desirability of alignment windows even with two interacting outages. Multiple near-optima at the edges of the calendar emerged (three practically equivalent alignments), which can be handled during shortlist/operations review.

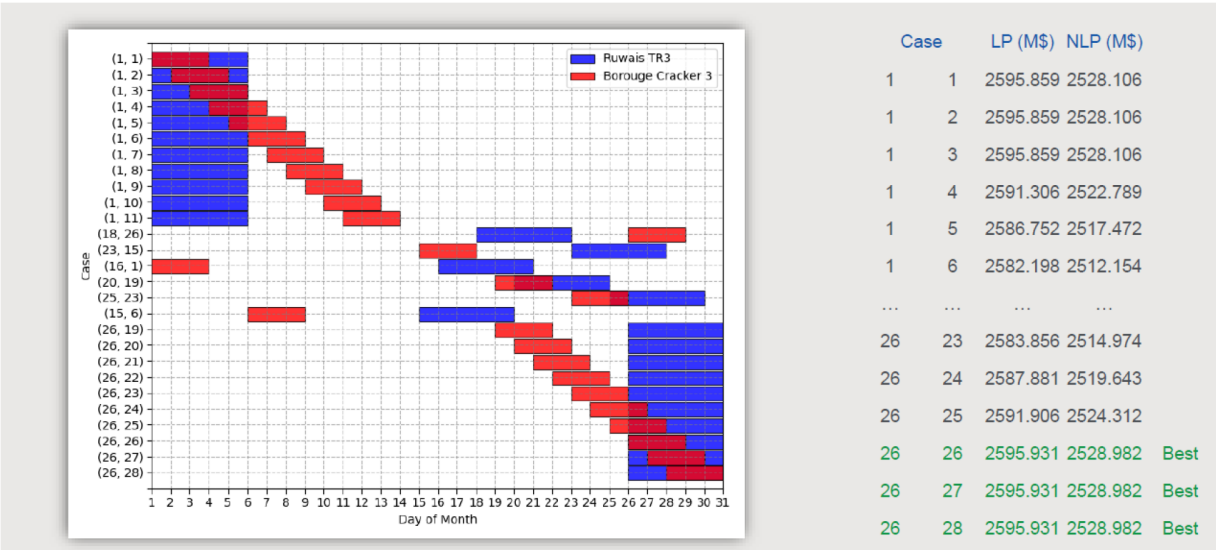


Figure 5—LP vs NLP on fixed double shutdown evaluation

Implication for VAST. These results support VAST's hybrid strategy: employ LP for fast, globally consistent exploration and use selective NLP polishing where composition fidelity is material (e.g., final checks on short-listed alignments). This balances tractability with rigor while maintaining solution repeatability for enterprise workflows.

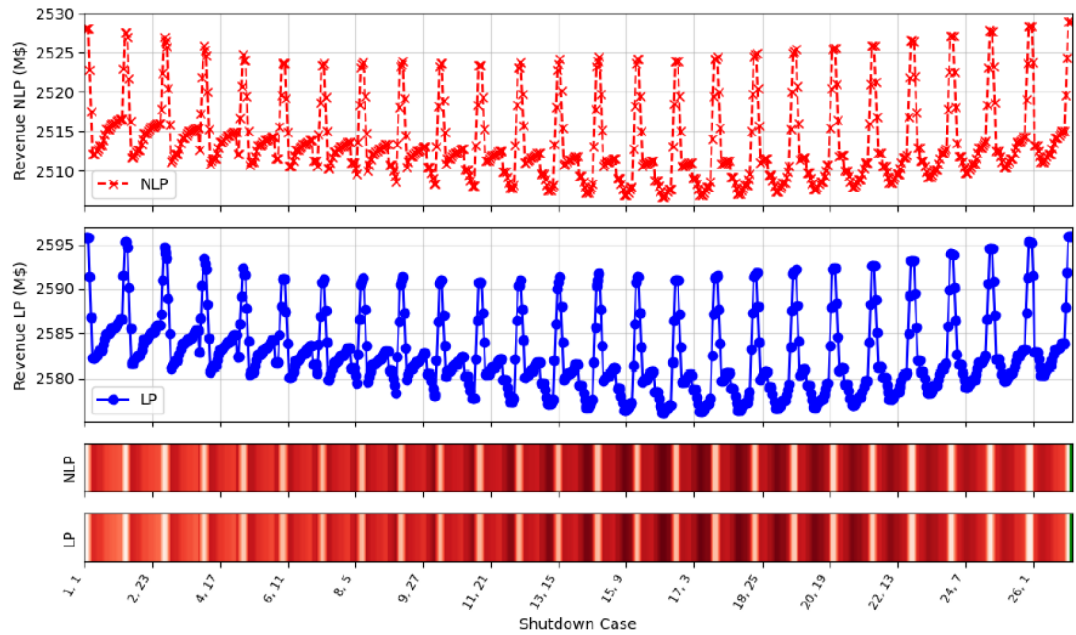


Figure 6—Shutdown alignment both in LP and NLP following the same sequence

On linearization fidelity and search stability. The LP-vs-NLP experiments demonstrated that a carefully crafted linearization reproduces the shape of the economic response with respect to shutdown timing and pairing, enabling the global search to find the same dates and alignment as NLP while reducing computational cost by two orders of magnitude across large candidate sets. In practice, this stabilizes GA convergence and allows broader sensitivity coverage within the same wall-clock budget.

Operationalization in annual planning. Because LP and NLP agree on the best windows (and on near-equivalent alternatives), the operating rhythm can confidently use LP for screening/shortlisting and reserve NLP checks for only the top candidates where compositional effects could influence final choice, minimizing friction with time-bound reviews and maximizing transparency for stakeholders.

Algorithmic trade-offs

The choice of optimization algorithms in VAST reflects a careful balance between computational efficiency, solution fidelity, and interpretability. Each approach—pure NLP, pure LP, and hybrid LP→NLP—offers distinct advantages and trade-offs, tailored to address the diverse requirements of shutdown planning in complex industrial networks.

Pure NLP provides high composition fidelity, making it ideal for modeling units with non-linear relationships between inputs and outputs. However, the trade-offs associated with NLP include susceptibility to local minima, which can lead to suboptimal solutions, and significantly longer runtimes due to the computational complexity of solving non-linear equations.

LP linearization simplifies the optimization problem by approximating non-linear relationships as linear, enabling faster exploration of the solution space. This approach is particularly effective for large-scale networks where computational efficiency is paramount, such as initial broad searches or rapid reporting requirements. LP accelerates the identification of promising shutdown combinations and facilitates scenario analysis, making it a valuable tool for high-level planning. However, the trade-off lies in reduced composition fidelity, as linearization may oversimplify critical flow dynamics, potentially overlooking nuanced operational constraints or economic impacts.

The hybrid approach uses LP during the alignment phase to efficiently explore shutdown combinations and identify promising solutions. Once the best alignment is found, NLP is applied to evaluate and refine the solution, ensuring compositionally rigorous checks for critical subproblems. This strategy balances computational efficiency with solution fidelity, enabling tractable optimization while preserving accuracy in key areas.

Algorithmic design and runtime

The evolution of VAST's algorithmic design reflects a continuous effort to enhance optimization performance, scalability, and solution quality. The system progressed through multiple versions, each introducing significant advancements in methodology and computational efficiency. Early versions (v1) utilized a Genetic Algorithm (GA) combined with Linear Programming (LP), this approach provided a solid foundation for rapid exploration but lacked the composition fidelity required for more complex scenarios.

Subsequent iterations (v2) incorporated Non-Linear Programming (NLP) into the optimization framework, enabling compositionally rigorous modeling for critical units. While this improved solution accuracy, it introduced challenges such as longer runtimes and increased sensitivity to local minima. Versions v3-v4 introduced analysis modes and breadth/depth hybrid algorithms, which balanced broad exploration with focused refinement, improving both runtime efficiency and solution quality. The latest version (v5) features a parallelized GA combined with deterministic polishing, leveraging multi-core processing to accelerate global search while ensuring precise day-level alignment of shutdown schedules.

Runtime performance is highly sensitive to the size of the network and the number of enabled shutdowns, with larger and more complex scenarios requiring greater computational resources. To address these challenges, parallelization and hardware scaling have been identified as effective levers for improving runtime efficiency.

Benchmarking among open-source solvers for full-network LP optimization indicated HiGHS as the preferred engine for this use case. HiGHS demonstrated superior performance in terms of runtime efficiency, scalability, and solution quality, making it an ideal choice for handling the linearized components of the optimization framework.

Architecture and interface

The VAST system has undergone significant evolution, transitioning from a constrained User Acceptance Testing (UAT) environment to a robust production deployment capable of handling the demands of large-scale industrial optimization. The production environment features enhanced computational resources, including increased CPU cores, RAM, and disk capacity, ensuring scalability and responsiveness for complex optimization tasks. Additionally, enterprise-grade Single Sign-On (SSO) has been integrated to streamline user authentication, providing secure and seamless access for authorized personnel. Role-based access control further enhances security by restricting functionality based on user roles, ensuring that sensitive data and critical operations are accessible only to designated users. Comprehensive system monitoring capabilities have also been implemented, enabling real-time tracking of system health, resource utilization, and job execution status.

The web-based interface is designed to provide an intuitive and user-friendly experience, offering a range of visualization and reporting tools to support decision-making. Key features include Gantt chart reporting for visualizing shutdown schedules, tabular reporting for detailed numerical analysis, and flowsheet visualization to represent network flows and compositions graphically. Users can compare results across multiple scenarios, enabling rapid evaluation of alternative shutdown combinations and their economic impacts. Full-result downloads allow users to export detailed optimization outputs for offline analysis or integration into external systems. Additionally, the system provides API access for automation, enabling seamless integration with enterprise workflows and external tools, such as data analytics platforms or scheduling software.

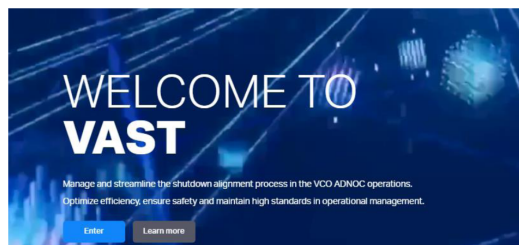


Figure 7—VAST accessing interface

Presentation of Data and Results

Solver selection, performance and repeatability

To determine the most suitable LP solver for VAST, two benchmark studies were conducted on ADNOC models. Since LP solvers return identical revenue for the same problem, the comparison focused on execution time and scalability.

In the May 2023 study, four model sizes were tested: AGP Asab, AGP Habshan, Gas Network, and Full Network. HiGHS consistently delivered the best performance, with approximately $2\times$ speed-up on Asab and $1.3\times$ on the full network compared to other open-source solvers.

In the February 2024 study, the full-network v4 model was benchmarked. HiGHS again showed superior performance with a mean time of 0.109 s and median of 0.091 s. Other solvers were slower: GLPK ($1.77\times$), CBC ($2.46\times$), Gurobi ($1.27\times$), and SCIP ($1.86\times$).

These results confirm HiGHS as the default LP solver for VAST, balancing speed and scalability. The tool remains solver-pluggable for future enhancements.

Table 1—Comparison between types of LP solvers

Solver	Mean Time (s)	Median Time (s)	Relative to HiGHS
HiGHS	0.109	0.091	1.00×
GLPK	0.168	0.161	1.77×
CBC	0.252	0.224	2.46×
Gurobi	0.150	0.116	1.27×
SCIP	0.174	0.169	1.86×

Full-network LP benchmarks showed HiGHS with the best mean/median times among tested engines. Repeatability is ensured via configurable random seeds and a deterministic post-processor, providing stable daily alignments while preserving global exploration.

Representative runtimes and scaling path

Illustrative end-to-end runtimes were approximately ~2 hours for ~72 enabled shutdowns and ~4 hours for ~175, in the tested environment. These runtimes reflect the system's ability to handle complex optimization tasks within a reasonable timeframe, balancing computational intensity with practical usability for industrial applications. However, as the scale and complexity of the optimization problem increases, further enhancements to computational performance are recommended to ensure responsiveness and scalability.

To achieve speed-ups and accommodate larger problem sizes, several scaling strategies are proposed. Vertical scaling through High- Performance Computing (HPC) resources, such as Azure H-series virtual machines, can significantly enhance processing power by leveraging advanced CPU architecture and increased memory capacity. This approach is particularly effective for scenarios requiring intensive computation, such as non-linear programming evaluations or large-scale genetic algorithm searches. Additionally, horizontal scaling through multi-instance deployments with orchestration can distribute workloads across multiple nodes, enabling parallel processing and reducing overall runtime. This scale-out strategy is well-suited for environments with fluctuating demand, allowing resources to be dynamically allocated based on workload requirements.

The optional integration of commercial solvers, such as Gurobi or CPLEX, offers another avenue for performance improvement. These solvers are optimized for handling large-scale linear and non-linear programming problems, providing faster convergence and enhanced solution quality. However, their adoption must be carefully evaluated against cost and complex trade-offs, as licensing fees and integration efforts may impact overall system feasibility.

User-facing outputs

The VAST application provides a suite of outputs designed to support decision-making and facilitate detailed analysis of shutdown planning scenarios. One of the key features is the generation of side-by-side Gantt charts that compare the original plan with the optimized shutdown schedule. These visualizations allow users to quickly identify improvements in timing, alignment, and resource utilization, highlighting the economic and operational benefits of the optimization process. Economic summaries are provided at both monthly and aggregated levels, offering a clear breakdown of revenue impacts, cost savings, and overall economic value generated by the optimized plan.

Streamflow and composition comparisons are another critical output, allowing users to analyze differences between scenarios or cases. These comparisons provide insights into how shutdown decisions affect material flows, product compositions, and network balances, enabling a deeper understanding of the trade-offs involved in each scenario. Diagnostic logs are generated for non-convergent subperiods, offering transparency into areas where the optimization process encountered challenges.

Flowsheet views further enhance the user experience by providing detailed visualizations of unit- and node-level balances. These views include clear unit systems and legends, ensuring that users can easily interpret the data and understand the relationships between different components of the network.



Figure 8—Dummy values on POC model for economic trends overall year.

IT and usability considerations

Production deployment required hardening (SSO, RBAC, CAB approvals), monitoring, and documentation. Training and adoption benefit from role-appropriate views, templated reports, and a clear annual operating rhythm (base-case setup, sensitivity matrix, shortlist, review, and share).

Conclusions

VAST demonstrates that AI-enabled global search combined with optimal-flow evaluation can operationalize shutdown alignment across an enterprise-scale hydrocarbon network. The deployed solution integrates validated models, benchmarked solvers, and a production-grade interface and API. UAT established functional completeness and model credibility, with targeted enhancements identified for speed and usability. The next steps planned include hardware scaling, broader user training, and continued refinement of input automation and reporting to support annual planning and "what-if" analyses. What used to take months and numerous number of people thanks to AI has been reduced to days with minimum number of people allowing efficiency and value maximization across ADNOC system.

Nomenclature

API	— Application Programming Interface
GA	— Genetic Algorithm
HPC	— High Performance Computing
HiGHS	— High-performance LP/MIP solver
LB	— Lower Bound
LP	— Linear Programming
NLP	— Non-Linear Programming
RBAC	— Role-Based Access Control
SSO	— Single Sign-On
UAT	— User Acceptance Testing
VAST	— VCO ADNOC Shutdown Tool
POC model	— Proof of Concept model representing a dummy configuration of a value chain

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